

Bala Divya Udagandla

Senior AI/ML Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer with **5+ years** of experience across financial services, MLOps, and telecommunications, specializing in building and deploying scalable, production-grade distributed machine learning systems. Engineered and automated end-to-end MLOps pipelines using **GitHub Actions and Terraform**, achieving over **75% reduction in model deployment time** for critical deep learning applications. Designed and implemented highly available, auto-scaling model serving infrastructure on **Kubernetes** to sustain unpredictable, high-throughput inference workloads without service degradation. Leveraged the full **AWS SageMaker** suite to streamline the end-to-end machine learning lifecycle, enhancing model experimentation velocity and reproducibility for data science teams. Collaborated closely with Data Science, Platform Engineering, and Product teams to define model requirements, operationalize complex deep learning solutions, and establish critical SLIs/SLOs. Holds **AWS Certified Machine Learning – Specialty** certification, demonstrating deep expertise in designing, implementing, and maintaining enterprise-grade ML solutions on the AWS cloud.

CORE COMPETENCIES

- MLOps & Automation | Distributed Model Training | Deep Learning Frameworks | Model Serving & Optimization
- Production ML Systems | High-Performance Computing | Data Engineering & Pipelines | Cloud Architecture (AWS)
- Containerization & Orchestration | Infrastructure as Code (IaC) | CI/CD for Machine Learning | System Design & Scalability

TECHNICAL SKILLS

Cloud Platforms: AWS (SageMaker, EKS, EC2, S3, Lambda, RDS, VPC), GCP (Vertex AI, GKE)

ML & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, PySpark MLlib, Hugging Face Transformers, Ray

Containers & Orchestration: Docker, Kubernetes (EKS), Helm, Kubeflow, Argo CD, Istio

Infrastructure as Code: Terraform, AWS CloudFormation, Ansible, Packer, Pulumi

CI/CD & DevOps: GitHub Actions, GitLab CI, Jenkins, CircleCI, Nexus, Artifactory

Monitoring & Observability: Prometheus, Grafana, Datadog, ELK Stack (Elasticsearch, Logstash, Kibana), AWS CloudWatch, W&B Platform

Programming & Scripting: Python (Pandas, NumPy, SciPy), Bash, SQL, Go, FastAPI, Flask

Databases & Streaming: PostgreSQL, MySQL, Redis, Apache Kafka, Apache Spark, RabbitMQ, S3, Snowflake

PROFESSIONAL EXPERIENCE

Nationwide-DELRAY beach,Florida

AI/ML Engineer | Jun 2024 – Present

Key Focus: MLOps & CI/CD • Generative AI & LLMs • Distributed Systems • Cloud-Native ML Infrastructure

- Engineered and deployed a scalable inference service for a new **Generative AI model** using **FastAPI** and **Kubernetes on EKS**, establishing a high-availability production endpoint.
- Developed a **CI/CD pipeline** with **GitHub Actions** and **Terraform** to automate model testing, containerization with **Docker**, and deployment, ensuring repeatable and reliable releases.
- Refactored a critical data preprocessing module in **PySpark**, significantly improving code maintainability and enhancing fault tolerance for upstream data sources.

- Designed and implemented a proof-of-concept for a **Retrieval-Augmented Generation (RAG)** system using **Lang Chain** and a **FAISS vector database** to improve contextual accuracy.
- Optimized an existing **XGBoost** model's **inference latency by 15%** through strategic feature engineering and hyperparameter tuning, directly impacting real-time risk assessment.
- Containerized a legacy machine learning application, facilitating its migration to a modern, **cloud-native MLOps platform** for improved scalability and observability.
- Established robust model monitoring practices using **Prometheus and Grafana** to track operational metrics, **model drift**, and data integrity SLOs.
- Contributed to the system design for a new **distributed training architecture** on **AWS SageMaker**, focusing on cost-efficiency and multi-node scalability for large-scale model development.

Technologies: Python | PyTorch | TensorFlow | Scikit-learn | AWS SageMaker | Kubernetes (EKS) | Docker | Terraform | GitHub Actions | FastAPI | PySpark | Prometheus

Weights & Biases- Hyderabad, India

AI/ML Engineer | Nov 2021 – Aug 2023

Key Focus: MLOps & System Design • Distributed Model Training • Scalable Inference • Model Lifecycle Management

- Architected and implemented a distributed training framework using **PyTorch DDP and Ray**, reducing large-scale model **training times by 35%** for internal research benchmarks.
- Engineered a scalable, real-time inference service on **Kubernetes (EKS)** with horizontal pod autoscaling, supporting a **5x increase** in prediction request throughput for a core product feature.
- Designed and deployed a comprehensive **MLOps pipeline** leveraging **W&B Artifacts** and **GitHub Actions** for automated model retraining, validation, and versioning.
- Pioneered the integration of advanced model explainability methods (**SHAP, Captum**) into the W&B platform, providing deeper insights for model debugging and analysis.
- Developed a **multi-modal transformer architecture** for a new feature prototype, establishing a new baseline for performance on complex, unstructured datasets.
- Led the design of a fault-tolerant data ingestion system using **Kafka and Spark Streaming** to reliably process high-volume experiment logs for analysis.
- Refactored monolithic model services into a scalable **microservices architecture**, improving system maintainability and enabling independent component deployment.
- Mentored junior engineers on MLOps best practices and system design principles for building reliable and scalable machine learning systems.
- Collaborated with product teams to define technical requirements and system specifications for new ML-powered features based on user research and feedback.

Technologies: Python | PyTorch | TensorFlow | Kubernetes (EKS) | Docker | W&B Platform | Ray | Spark | Kafka | Scikit-learn | AWS

Ericsson- Hyderabad, India

Python Developer | Jul 2020 – Nov 2021

Key Focus: Predictive Analytics • MLOps & Deployment • Big Data Engineering • Time-Series Forecasting

- Engineered a time-series forecasting model using **LSTM networks in TensorFlow** to predict network traffic surges, improving forecast **accuracy by 18%** over the legacy ARIMA-based system.
- Designed and deployed a scalable model serving infrastructure on **AWS SageMaker**, establishing a **CI/CD pipeline** for automated model retraining and deployment.
- Implemented a predictive maintenance model with **XGBoost** to identify telecom hardware at risk of failure, enabling proactive scheduling of engineering field support.
- Architected and automated robust data processing pipelines using **Apache Spark and Airflow** to ETL terabytes of raw network telemetry data for model training.
- Developed a **BERT-based NLP model** to automatically classify and route technical support tickets from network operations centers, improving issue categorization.
- Containerized machine learning applications using **Docker** and orchestrated their deployment on a **Kubernetes cluster (EKS)** for high availability and fault tolerance.
- Conducted feature engineering and selection from complex, high-dimensional network data streams to enhance the predictive power of anomaly detection models.

- Refactored monolithic Python ML codebases into a microservices architecture, deploying them as serverless functions with **AWS Lambda and API Gateway** for enhanced scalability.
- Collaborated with network engineering teams to define service level objectives (SLOs) for ML-driven network optimization and resource allocation services.

Technologies: Python | TensorFlow | PyTorch | scikit-learn | Apache Spark | AWS SageMaker | Kubernetes | Docker | Airflow | XGBoost

PROJECTS

RAG Inference Engine | github.com/Divya200145/rag-inference-engine

Python • LangChain • FAISS • FastAPI • Docker • Kubernetes

- Built a production-grade **Retrieval-Augmented Generation (RAG)** pipeline with document ingestion, chunking, embedding, and **FAISS**-based semantic retrieval, served through a **FastAPI** inference endpoint.
- Containerized the service with **Docker** and deployed on **Kubernetes** with health checks and horizontal autoscaling for high-availability, low-latency contextual responses.

Distributed Training Framework | github.com/Divya200145/distributed-training-framework

Python • PyTorch DDP • Ray • AWS • MLflow

- Developed a multi-node, multi-GPU training framework using **PyTorch DistributedDataParallel (DDP)** and **Ray**, with fault-tolerant checkpointing and automatic experiment tracking.
- Implemented gradient accumulation, mixed-precision training, and dynamic resource scheduling, reducing large-scale training times by **35%** on benchmark workloads.

Real-Time ML Serving Platform | github.com/Divya200145/realtime-ml-serving

Python • FastAPI • Kubernetes (EKS) • Prometheus • Grafana

- Engineered a low-latency model serving platform on **Kubernetes (EKS)** with horizontal pod autoscaling, supporting a **5x increase** in prediction request throughput.
- Instrumented end-to-end observability with **Prometheus** and **Grafana** dashboards tracking latency, throughput, model drift, and data-integrity SLOs.

EDUCATION

Master's Degree (COMPUTER Science)

Florida Atlantic University • 2025

CERTIFICATIONS

- Azure AI Fundamentals (AI-901)
- AWS Certified AI Practitioner
- Big Data Analytics